

6.3 A Generic Adaptive Equalizer

An adaptive equalizer is a time-varying filter which must constantly be retuned. The basic structure of an adaptive equalizer is shown in Figure 6.2, where the subscript k is used to denote a discrete time index (in section 6.4, another notation is introduced to represent discrete time events — both notations have exactly the same meaning).

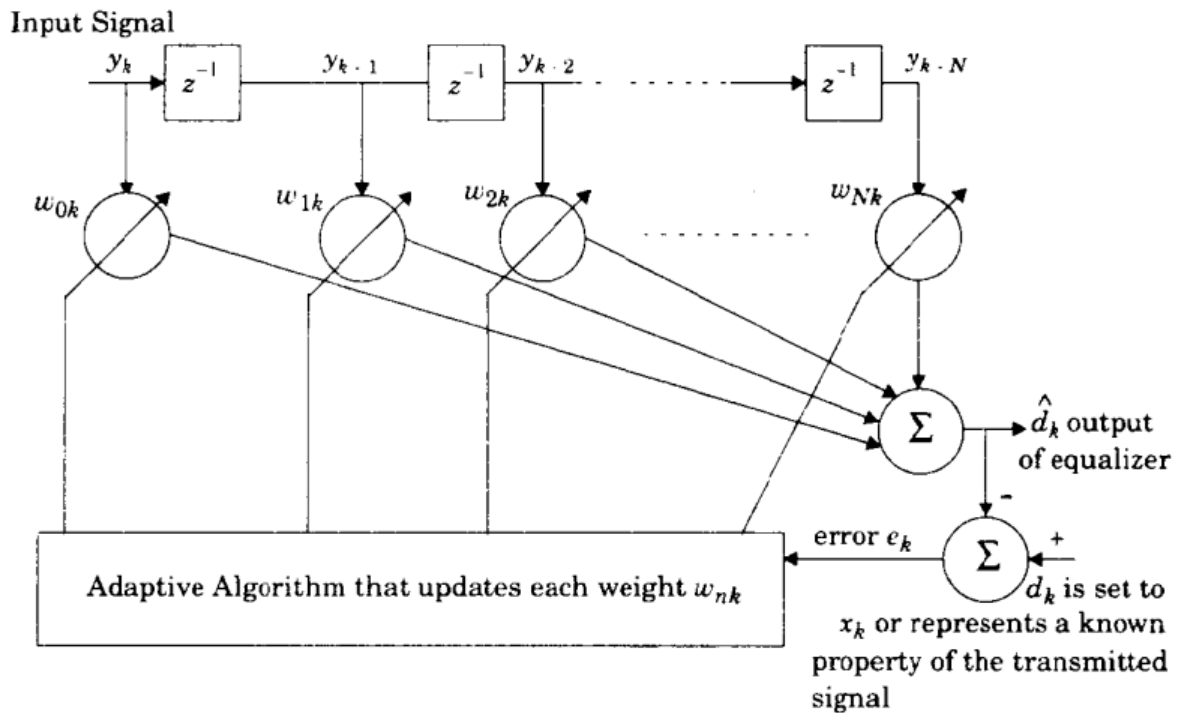


Figure 6.2
A basic linear equalizer during training.

Notice in Figure 6.2 that there is a single input y_k at any time instant. The value of y_k depends upon the instantaneous state of the radio channel and the specific value of the noise (see Figure 6.1). As such, y_k is a random process. The adaptive equalizer structure shown above is called a *transversal filter*, and in this case has N delay elements, $N + 1$ taps, and $N + 1$ tunable complex multipliers, called *weights*. The weights of the filter are described by their physical location in the delay line structure, and have a second subscript, k , to explicitly show they vary with time. These weights are updated continuously by the adaptive

algorithm, either on a sample by sample basis (i.e., whenever k is incremented by 1) or on a block by block basis (i.e., whenever a specified number of samples have been clocked into the equalizer).

The adaptive algorithm is controlled by the error signal e_k . This error signal is derived by comparing the output of the equalizer, $\hat{d}_k$, with some signal d_k which is either an exact scaled replica of the transmitted signal x_k or which represents a known property of the transmitted signal. The adaptive algorithm uses e_k to minimize a *cost function* and updates the equalizer weights in a manner that iteratively reduces the cost function. For example, the least mean squares (LMS) algorithm searches for the optimum or near-optimum filter weights by performing the following iterative operation:

$$\text{New weights} = \text{Previous weights} + (\text{constant}) \times (\text{Previous error}) \times (\text{Current input vector}) \quad (6.6.a)$$

where

$$\text{Previous error} = \text{Previous desired output} - \text{Previous actual output} \quad (6.6.b)$$

and the constant may be adjusted by the algorithm to control the variation between filter weights on successive iterations. This process is repeated rapidly in a programming loop while the equalizer attempts to *converge*, and many techniques (such as gradient or steepest decent algorithms) may be used to minimize the error. Upon reaching convergence, the adaptive algorithm freezes the filter weights until the error signal exceeds an acceptable level or until a new training sequence is sent.

Based on classical equalization theory [Wid85], [Qur85], the most common cost function is the mean square error (MSE) between the desired signal and the output of the equalizer. The MSE is denoted by $E[e(k)e^*(k)]$, and when a replica of the transmitted signal is required at the output of the equalizer (i.e., when d_k is set equal to x_k and is known *a priori*), a known *training sequence* must be periodically transmitted. By detecting the training sequence, the adaptive algorithm in the receiver is able to compute and minimize the cost function by driving the tap weights until the next training sequence is sent.

A more recent class of adaptive algorithms are able to exploit characteristics of the transmitted signal and do not require training sequences. These modern algorithms are able to acquire equalization through property restoral techniques of the transmitted signal, and are called *blind algorithms* because they provide equalizer convergence without burdening the transmitter with training overhead. These techniques include algorithms such as the *constant modulus algorithm* (CMA) and the *spectral coherence restoral algorithm* (SCORE). CMA is used for constant envelope modulation, and forces the equalizer weights to maintain a constant envelope on the received signal [Tre83], whereas SCORE exploits spectral redundancy or cyclostationarity in the transmitted signal [Gar91]. Blind algorithms are not described in this text, but are becoming important for wireless applications.

To study the adaptive equalizer of Figure 6.2, it is helpful to use vector and matrix algebra. Define the input signal to the equalizer as a vector $\mathbf{y}_k$ where

$$\mathbf{y}_k = [y_k \quad y_{k-1} \quad y_{k-2} \quad \dots \quad y_{k-N}]^T \quad (6.7)$$

It should be clear that the output of the adaptive equalizer is a scalar given by

$$\hat{d}_k = \sum_{n=0}^N w_{nk} y_{k-n} \quad (6.8)$$

and following equation (6.7) a weight vector can be written as

$$\mathbf{w}_k = [w_{0k} \quad w_{1k} \quad w_{2k} \quad \dots \quad w_{Nk}]^T \quad (6.9)$$

Using (6.7) and (6.9), (6.8) may be written in vector notation as

$$\hat{d}_k = \mathbf{y}_k^T \mathbf{w}_k = \mathbf{w}_k^T \mathbf{y}_k \quad (6.10)$$

It follows that when the desired equalizer output is known (i.e., $d_k = x_k$), the error signal e_k is given by

$$e_k = d_k - \hat{d}_k = x_k - \hat{d}_k \quad (6.11)$$

and from (6.10)

$$e_k = x_k - \mathbf{y}_k^T \mathbf{w}_k = x_k - \mathbf{w}_k^T \mathbf{y}_k \quad (6.12)$$

To compute the mean square error $|e_k|^2$ at time instant k , equation (6.12) is squared to obtain

$$|e_k|^2 = x_k^2 + \mathbf{w}_k^T \mathbf{y}_k \mathbf{y}_k^T \mathbf{w}_k - 2x_k \mathbf{y}_k^T \mathbf{w}_k \quad (6.13)$$

Taking the expected value of $|e_k|^2$ over k (which in practice amounts to computing a time average) yields

$$E[|e_k|^2] = E[x_k^2] + \mathbf{w}_k^T E[\mathbf{y}_k \mathbf{y}_k^T] \mathbf{w}_k - 2E[x_k \mathbf{y}_k^T] \mathbf{w}_k \quad (6.14)$$

Notice that the filter weights $\mathbf{w}_k$ are not included in the time average since, for convenience, it is assumed that they have converged to the optimum value and are not varying with time.

Equation (6.14) would be trivial to simplify if x_k and y_k were independent. However, this is not true in general since the input vector should be correlated with the desired output of the equalizer (otherwise, the equalizer would have a very difficult time extracting the desired signal). Instead, the cross correlation vector $\mathbf{p}$ between the desired response and the input signal is defined as

$$\mathbf{p} = E[x_k \mathbf{y}_k] = E \begin{bmatrix} x_k y_k & x_k y_{k-1} & x_k y_{k-2} & \dots & x_k y_{k-N} \end{bmatrix}^T \quad (6.15)$$

and the *input correlation matrix* is defined as the $(N+1) \times (N+1)$ square matrix $\mathbf{R}$ where

$$\mathbf{R} = E[\mathbf{y}_k \mathbf{y}_k^*] = E \begin{bmatrix} y_k^2 & y_k y_{k-1} & y_k y_{k-2} & \dots & y_k y_{k-N} \\ y_{k-1} y_k & y_{k-1}^2 & y_{k-1} y_{k-2} & \dots & y_{k-1} y_{k-N} \\ \dots & \dots & \dots & \dots & \dots \\ y_{k-N} y_k & y_{k-N} y_{k-1} & y_{k-N} y_{k-2} & \dots & y_{k-N}^2 \end{bmatrix} \quad (6.16)$$

The matrix $\mathbf{R}$ is sometimes called the *input covariance matrix*. The major diagonal of $\mathbf{R}$ contains the mean square values of each input sample, and the cross terms specify the autocorrelation terms resulting from delayed samples of the input signal.

If x_k and y_k are stationary, then the elements in $\mathbf{R}$ and $\mathbf{p}$ are second order statistics which do not vary with time. Using equations (6.15) and (6.16), (6.14) may be rewritten as

$$\text{Mean Square Error} \equiv \xi = E[x_k^2] + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{p}^T \mathbf{w} \quad (6.17)$$

By minimizing equation (6.17) in terms of the weight vector $\mathbf{w}_k$, it becomes possible to adaptively tune the equalizer to provide a flat spectral response (minimal ISI) in the received signal. This is due to the fact that when the input signal y_k and the desired response x_k are stationary, the mean square error (MSE) is quadratic on $\mathbf{w}_k$, and minimizing the MSE leads to optimal solutions for $\mathbf{w}_k$.

Example 6.1

The MSE of equation (6.17) is a multidimensional function. When two tap weights are used in an equalizer, then the MSE function is a bowl-shaped paraboloid where MSE is plotted on the vertical axis and weights w_0 and w_1 are plotted on the horizontal axes. If more than two tap-weights are used in the equalizer, then the error function is a hyperparaboloid. In all cases the error function is concave upward, which implies that a minimum may be found [Wid85].

To determine the minimum MSE (MMSE), the gradient of (6.17) can be used. As long as $\mathbf{R}$ is nonsingular (has an inverse), the MMSE occurs when $\mathbf{w}_k$ are such that the gradient is zero. The gradient of ξ is defined as

$$\nabla \equiv \frac{\partial \xi}{\partial \mathbf{w}} = \left[\frac{\partial \xi}{\partial w_0} \quad \frac{\partial \xi}{\partial w_1} \quad \dots \quad \frac{\partial \xi}{\partial w_N} \right]^T \quad (\text{E6.1})$$

By expanding (6.17) and differentiating with respect to each signal in the weight vector, it can be shown that equation (E6.1) yields

$$\nabla = 2\mathbf{R}\mathbf{w} - 2\mathbf{p} \quad (\text{E6.2})$$

Setting $\nabla = 0$ in (E6.2), the optimum weight vector $\hat{\mathbf{w}}$ for MMSE is given by

$$\hat{\mathbf{w}} = \mathbf{R}^{-1} \mathbf{p} \quad (\text{E6.3})$$

Example 6.2

Four matrix algebra rules are useful in the study of adaptive equalizers [Wid85]:

1. Differentiation of $\mathbf{w}^T \mathbf{R} \mathbf{w}$ can be differentiated as the product $(\mathbf{w}^T) (\mathbf{R} \mathbf{w})$.
2. For any square matrix, $\mathbf{A} \mathbf{A}^{-1} = \mathbf{I}$.
3. For any matrix product, $(\mathbf{A} \mathbf{B})^T = \mathbf{B}^T \mathbf{A}^T$.
4. For any symmetric matrix, $\mathbf{A}^T = \mathbf{A}$ and $(\mathbf{A}^{-1})^T = \mathbf{A}^{-1}$.

Using (E6.3) to substitute $\hat{\mathbf{w}}$ for $\mathbf{w}$ in equation (6.17), and using the above rules, ξ_{min} is found to be

$$\xi_{min} = \text{MMSE} = E[x_k^2] - \mathbf{p}^T \mathbf{R}^{-1} \mathbf{p} = E[x_k^2] - \mathbf{p}^T \hat{\mathbf{w}} \quad (\text{E6.4})$$

Equation (E6.4) solves the MMSE for optimum weights.

9.3 DIGITAL SUBSCRIBER LINE

After traditional modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet. Digital subscriber line (DSL) technology is one of the most promising for supporting high-speed digital communication over the existing local loops. DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL). The set is often referred to as xDSL, where x can be replaced by A, V, H, or S.

ADSL

The first technology in the set is asymmetric DSL (ADSL). ADSL, like a 56K modem, provides higher speed (bit rate) in the downstream direction (from the Internet to the resident) than in the upstream direction (from the resident to the Internet). That is the reason it is called asymmetric. Unlike the asymmetry in 56K modems, the designers of ADSL specifically divided the available bandwidth of the local loop unevenly for the residential customer. The service is not suitable for business customers who need a large bandwidth in both directions.

ADSL is an asymmetric communication technology designed for residential users; it is not suitable for businesses.

Using Existing Local Loops

One interesting point is that ADSL uses the existing local loops. But how does ADSL reach a data rate that was never achieved with traditional modems? The answer is that the twisted-pair local loop is actually capable of handling bandwidths up to 1.1 MHz, but the filter installed at the end office of the telephone company where each local loop terminates limits the bandwidth to 4 kHz (sufficient for voice communication). If the filter is removed, however, the entire 1.1 MHz is available for data and voice communications.

The existing local loops can handle bandwidths up to 1.1 MHz.

Adaptive Technology

Unfortunately, 1.1 MHz is just the theoretical bandwidth of the local loop. Factors such as the distance between the residence and the switching office, the size of the cable, the signaling used, and so on affect the bandwidth. The designers of ADSL technology were aware of this problem and used an adaptive technology that tests the condition and bandwidth availability of the line before settling on a data rate. The data rate of ADSL is not fixed; it changes based on the condition and type of the local loop cable.

ADSL is an adaptive technology. The system uses a data rate based on the condition of the local loop line.

Discrete Multitone Technique

The modulation technique that has become standard for ADSL is called the discrete multitone technique (DMT) which combines QAM and FDM. There is no set way that the bandwidth of a system is divided. Each system can decide on its bandwidth division. Typically, an available bandwidth of 1.104 MHz is divided into 256 channels. Each channel uses a bandwidth of 4.312 kHz, as shown in Figure 9.10. Figure 9.11 shows how the bandwidth can be divided into the following:

- Voice. Channel 0 is reserved for voice communication.
- Idle. Channels 1 to 5 are not used and provide a gap between voice and data communication.

Figure 9.10 Discrete multitone technique

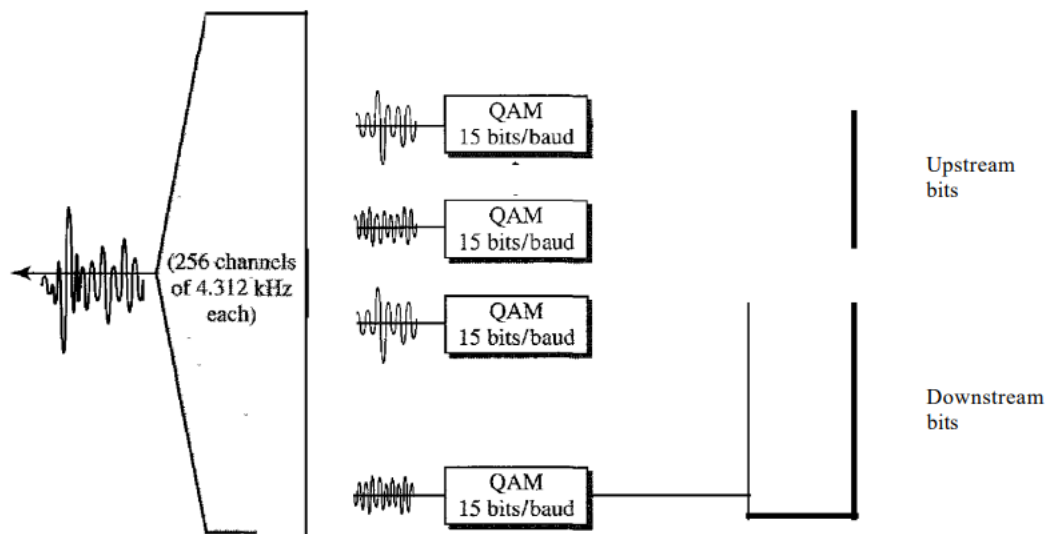
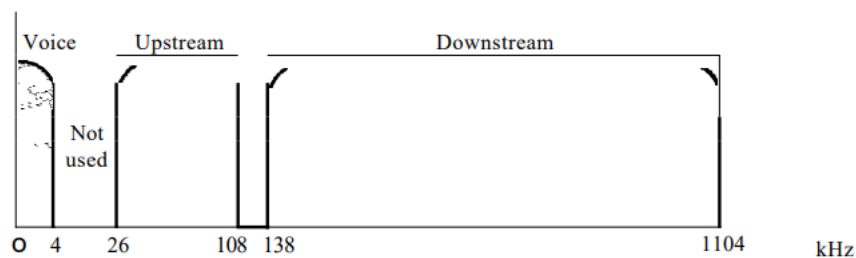


Figure 9.11 Bandwidth division in ADSL

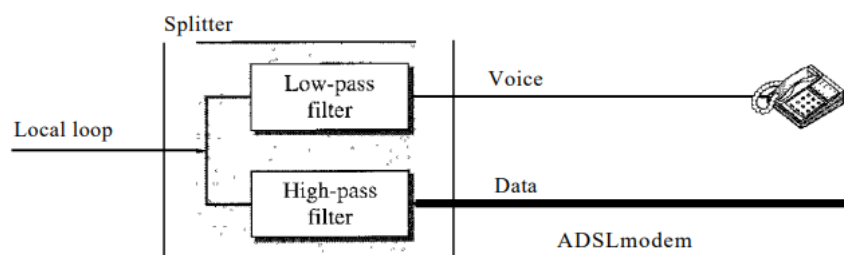


- Upstream data and control. Channels 6 to 30 (25 channels) are used for upstream data transfer and control. One channel is for control, and 24 channels are for data transfer. If there are 24 channels, each using 4 kHz (out of 4.312 kHz available) with QAM modulation, we have $24 \times 4000 \times 15$, or a 1.44-Mbps bandwidth, in the upstream direction. However, the data rate is normally below 500 kbps because some of the carriers are deleted at frequencies where the noise level is large. In other words, some of channels may be unused.
- Downstream data and control. Channels 31 to 255 (225 channels) are used for downstream data transfer and control. One channel is for control, and 224 channels are for data. If there are 224 channels, we can achieve up to $224 \times 4000 \times 15$, or 13.4 Mbps. However, the data rate is normally below 8 Mbps because some of the carriers are deleted at frequencies where the noise level is large. In other words, some of channels may be unused.

Customer Site: ADSL Modem

Figure 9.12 shows an ADSL modem installed at a customer's site. The local loop connects to a splitter which separates voice and data communications. The ADSL modem modulates and demodulates the data, using DMT, and creates downstream and upstream channels.

Figure 9.12 *ADSL modem*

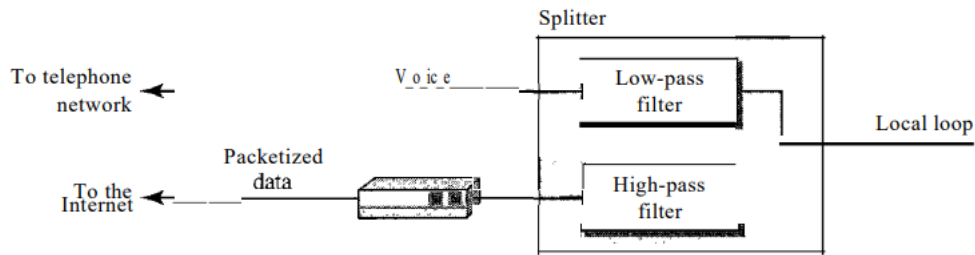


Note that the splitter needs to be installed at the customer's premises, normally by a technician from the telephone company. The voice line can use the existing telephone wiring in the house, but the data line needs to be installed by a professional. All this makes the ADSL line expensive. We will see that there is an alternative technology, Universal ADSL (or ADSL Lite).

Telephone Company Site: DSLAM

At the telephone company site, the situation is different. Instead of an ADSL modem, a device called a digital subscriber line access multiplexer (DSLAM) is installed that functions similarly. In addition, it packetizes the data to be sent to the Internet (ISP server). Figure 9.13 shows the configuration.

Figure 9.13 *DSLAM*



ADSL Lite

The installation of splitters at the border of the premises and the new wiring for the data line can be expensive and impractical enough to dissuade most subscribers. A new version of ADSL technology called ADSL Lite (or Universal ADSL or splitterless ADSL) is available for these subscribers. This technology allows an ADSL Lite modem to be plugged directly into a telephone jack and connected to the computer. The splitting is done at the telephone company. ADSL Lite uses 256 DMT carriers with 8-bit modulation (instead of 15-bit). However, some of the carriers may not be available because errors created by the voice signal might mingle with them. It can provide a maximum downstream data rate of 1.5 Mbps and an upstream data rate of 512 kbps.

HDSL

The high-bit-rate digital subscriber line (HDSL) was designed as an alternative to the T-1 line (1.544 Mbps). The T-1 line uses alternate mark inversion (AMI) encoding, which is very susceptible to attenuation at high frequencies. This limits the length of a T-1 line to 3200 ft (1 km). For longer distances, a repeater is necessary, which means increased costs.

HDSL uses 2B1Q encoding (see Chapter 4), which is less susceptible to attenuation. A data rate of 1.544 Mbps (sometimes up to 2 Mbps) can be achieved without repeaters up to a distance of 12,000 ft (3.86 km). HDSL uses two twisted pairs (one pair for each direction) to achieve full-duplex transmission.

SDSL

The symmetric digital subscriber line (SDSL) is a one twisted-pair version of HDSL. It provides full-duplex symmetric communication supporting up to 768 kbps in each direction. SDSL, which provides symmetric communication, can be considered an alternative to ADSL. ADSL provides asymmetric communication, with a downstream bit rate that is much higher than the upstream bit rate. Although this feature meets the needs of most residential subscribers, it is not suitable for businesses that send and receive data in large volumes in both directions.

VDSL

The very high-bit-rate digital subscriber line (VDSL), an alternative approach that is similar to ADSL, uses coaxial, fiber-optic, or twisted-pair cable for short distances. The modulating technique is DMT. It provides a range of bit rates (25 to 55 Mbps) for upstream communication at distances of 3000 to 10,000 ft. The downstream rate is normally 3.2 Mbps.

Summary

Table 9.1 shows a summary of DSL technologies. Note that the data rate and distances are approximations and can vary from one implementation to another.

Table 9.1 *Summary of DSL technologies*

<i>Technology</i>	<i>Downstream Rate</i>	<i>Upstream Rate</i>	<i>Distance (ft)</i>	<i>Twisted Pairs</i>	<i>Line Code</i>
ADSL	1.5-6.1 Mbps	16-640 kbps	12,000	1	DMT
ADSL Lite	1.5 Mbps	500 kbps	18,000	1	DMT
HDSL	1.5-2.0 Mbps	1.5-2.0 Mbps	12,000	2	2B1Q
SDSL	768 kbps	768 kbps	12,000	1	2B1Q
VDSL	25-55 Mbps	3.2 Mbps	3000-10,000	1	DMT

Small-scale fading, or simply *fading*, is used to describe the rapid fluctuation of the amplitude of a radio signal over a short period of time or travel distance, so that large-scale path loss effects may be ignored. Fading is caused by interference between two or more versions of the transmitted signal which arrive at the receiver at slightly different times. These waves, called *multipath waves*, combine at the receiver antenna to give a resultant signal which can vary widely in amplitude and phase, depending on the distribution of the intensity and relative propagation time of the waves and the bandwidth of the transmitted signal.

4.1 Small-Scale Multipath Propagation

Multipath in the radio channel creates small-scale fading effects. The three most important effects are:

- Rapid changes in signal strength over a small travel distance or time interval
- Random frequency modulation due to varying Doppler shifts on different multipath signals
- Time dispersion (echoes) caused by multipath propagation delays.

In built-up urban areas, fading occurs because the height of the mobile antennas are well below the height of surrounding structures, so there is no single line-of-sight path to the base station. Even when a line-of-sight exists, multipath still occurs due to reflections from the ground and surrounding structures. The incoming radio waves arrive from different directions with different propagation delays. The signal received by the mobile at any point in space may consist of a large number of plane waves having randomly distributed amplitudes,

phases, and angles of arrival. These multipath components combine vectorially at the receiver antenna, and can cause the signal received by the mobile to distort or fade. Even when a mobile receiver is stationary, the received signal may fade due to movement of surrounding objects in the radio channel.

If objects in the radio channel are static, and motion is considered to be only due to that of the mobile, then fading is purely a spatial phenomenon. The spatial variations of the resulting signal are seen as temporal variations by the receiver as it moves through the multipath field. Due to the constructive and destructive effects of multipath waves summing at various points in space, a receiver moving at high speed can pass through several fades in a small period of time. In a more serious case, a receiver may stop at a particular location at which the received signal is in a deep fade. Maintaining good communications can then become very difficult, although passing vehicles or people walking in the vicinity of the mobile can often disturb the field pattern, thereby diminishing the likelihood of the received signal remaining in a deep null for a long period of time. Antenna space diversity can prevent deep fading nulls, as shown in Chapter 6. Figure 3.1 shows typical rapid variations in the received signal level due to small-scale fading as a receiver is moved over a distance of a few meters.

Due to the relative motion between the mobile and the base station, each multipath wave experiences an apparent shift in frequency. The shift in received signal frequency due to motion is called the Doppler shift, and is directly proportional to the velocity and direction of motion of the mobile with respect to the direction of arrival of the received multipath wave.

4.1.1 Factors Influencing Small-Scale Fading

Many physical factors in the radio propagation channel influence small-scale fading. These include the following:

- **Multipath propagation** — The presence of reflecting objects and scatterers in the channel creates a constantly changing environment that dissipates the signal energy in amplitude, phase, and time. These effects result in multiple versions of the transmitted signal that arrive at the receiving antenna, displaced with respect to one another in time and spatial orientation. The random phase and amplitudes of the different multipath components cause fluctuations in signal strength, thereby inducing small-scale fading, signal distortion, or both. Multipath propagation often lengthens the time required for the baseband portion of the signal to reach the receiver which can cause signal smearing due to intersymbol interference.
- **Speed of the mobile** — The relative motion between the base station and the mobile results in random frequency modulation due to different Doppler shifts on each of the multipath components. Doppler shift will be positive or negative depending on whether the mobile receiver is moving toward or away from the base station.
- **Speed of surrounding objects** — If objects in the radio channel are in motion, they induce a time varying Doppler shift on multipath components. If the surrounding objects move at a greater rate than the mobile, then this effect dominates the small-scale fading. Otherwise, motion of surrounding objects may be ignored, and only the speed of the mobile need be considered.
- **The transmission bandwidth of the signal** — If the transmitted radio signal bandwidth is greater than the “bandwidth” of the multipath channel, the received signal will be distorted, but the received signal strength will not fade much over a local area (i.e., the small-scale signal fading will not be significant). As will be shown, the bandwidth of the channel can be quantified by the *coherence bandwidth* which is related to the specific multipath structure of the channel. The coherence bandwidth is a measure of the maximum frequency difference for which signals are still strongly correlated in amplitude. If the transmitted signal has a narrow bandwidth as compared to the channel, the amplitude of the signal will change rapidly, but the signal will not be distorted in time. Thus, the statistics of small-scale signal strength and the likelihood of signal smearing appearing over small-scale distances are very much related to the specific amplitudes and delays of the multipath channel, as well as the bandwidth of the transmitted signal.

4.6.1 Rayleigh Fading Distribution

In mobile radio channels, the Rayleigh distribution is commonly used to describe the statistical time varying nature of the received envelope of a flat fading signal, or the envelope of an individual multipath component. It is well known that the envelope of the sum of two quadrature Gaussian noise signals obeys a Rayleigh distribution. Figure 4.15 shows a Rayleigh distributed signal envelope as a function of time. The Rayleigh distribution has a probability density function (pdf) given by

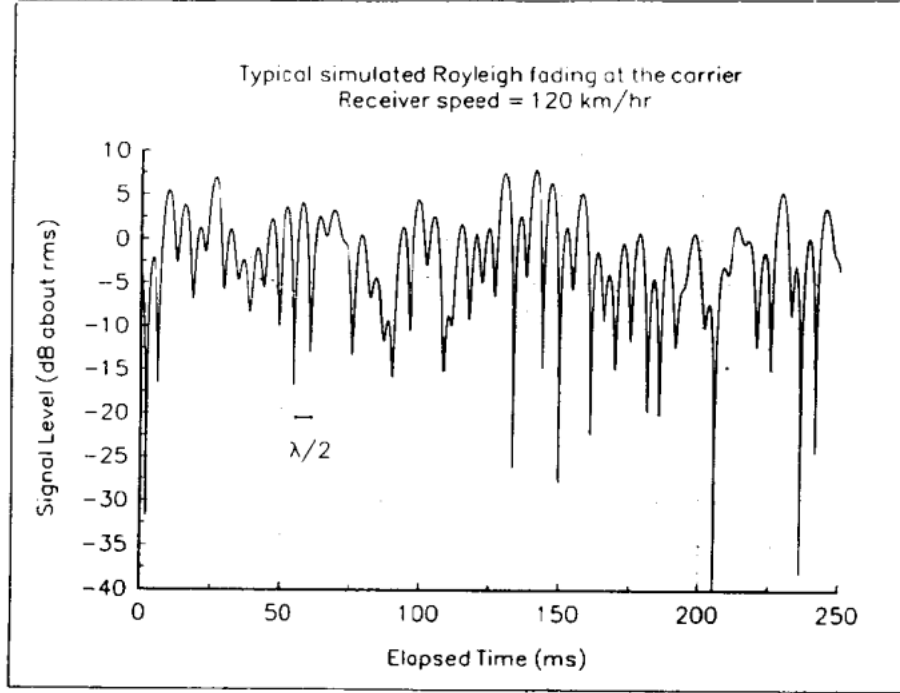


Figure 4.15

A typical Rayleigh fading envelope at 900 MHz [From [Fun93] © IEEE].

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & (0 \leq r \leq \infty) \\ 0 & (r < 0) \end{cases} \quad (4.49)$$

where σ is the rms value of the received voltage signal before *envelope detection*, and σ^2 is the time-average power of the received signal *before envelope detection*. The probability that the envelope of the received signal does not exceed a specified value R is given by the corresponding cumulative distribution function (CDF)

$$P(R) = Pr(r \leq R) = \int_0^R p(r) dr = 1 - \exp\left(-\frac{R^2}{2\sigma^2}\right) \quad (4.50)$$

The mean value r_{mean} of the Rayleigh distribution is given by

$$r_{mean} = E[r] = \int_0^\infty r p(r) dr = \sigma \sqrt{\frac{\pi}{2}} = 1.2533\sigma \quad (4.51)$$

and the variance of the Rayleigh distribution is given by σ_r^2 , which represents the ac power in the signal envelope

$$\begin{aligned}\sigma_r^2 &= E[r^2] - E^2[r] = \int_0^{\infty} r^2 p(r) dr - \frac{\sigma^2 \pi}{2} \\ &= \sigma^2 \left(2 - \frac{\pi}{2} \right) = 0.4292 \sigma^2\end{aligned}\quad (4.52)$$

The rms value of the envelope is the square root of the mean square, or $\sqrt{2}\sigma$.

The median value of r is found by solving

$$\frac{1}{2} = \int_0^{r_{\text{median}}} p(r) dr \quad (4.53)$$

and is

$$r_{\text{median}} = 1.177\sigma \quad (4.54)$$

Thus the mean and the median differ by only 0.55 dB in a Rayleigh fading signal. Note that the median is often used in practice, since fading data are usually measured in the field and a particular distribution cannot be assumed. By using median values instead of mean values it is easy to compare different fading distributions which may have widely varying means. Figure 4.16 illustrates the Rayleigh pdf. The corresponding Rayleigh cumulative distribution function (CDF) is shown in Figure 4.17.

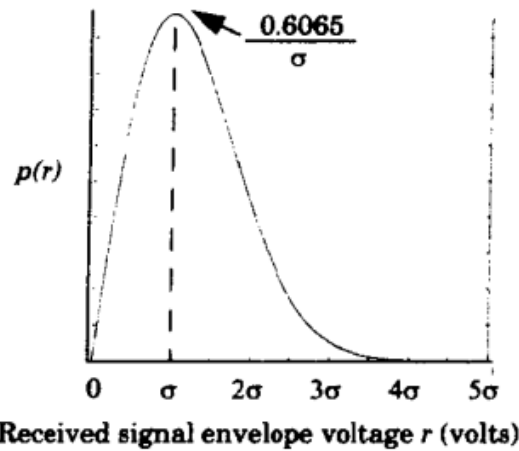


Figure 4.16
Rayleigh probability density function (pdf).

Jakes's model [\[edit\]](#)

In his book,^[6] Jakes popularised a model for Rayleigh fading based on summing [sinusoids](#). Let the scatterers be uniformly distributed around a circle at angles α_n with k rays emerging from each scatterer. The Doppler shift on ray n is

$$f_n = f_d \cos \alpha_n$$

and, with M such scatterers, the Rayleigh fading of the k^{th} waveform over time t can be modelled as:

$$R(t, k) = 2\sqrt{2} \left[\sum_{n=1}^M (\cos \beta_n + j \sin \beta_n) \cos(2\pi f_n t + \theta_{n,k}) + \frac{1}{\sqrt{2}} (\cos \alpha + j \sin \alpha) \cos(2\pi f_d t) \right].$$

Here, α and the β_n and $\theta_{n,k}$ are model parameters with α usually set to zero, β_n chosen so that there is no cross-correlation between the real and imaginary parts of $R(t)$:

$$\beta_n = \frac{\pi n}{M+1}$$

and $\theta_{n,k}$ used to generate multiple waveforms. If a single-path channel is being modelled, so that there is only one waveform then θ_n can be zero. If a multipath, frequency-selective channel is being modelled so that multiple waveforms are needed, Jakes suggests that uncorrelated waveforms are given by

$$\theta_{n,k} = \beta_n + \frac{2\pi(k-1)}{M+1}.$$

In fact, it has been shown that the waveforms are correlated among themselves — they have non-zero cross-correlation — except in special circumstances.^[7] The model is also [deterministic](#) (it has no random element to it once the parameters are chosen). A modified Jakes's model^[8] chooses slightly different spacings for the scatterers and scales their waveforms using [Walsh–Hadamard sequences](#) to ensure zero cross-correlation. Setting

$$\alpha_n = \frac{\pi(n-0.5)}{2M} \text{ and } \beta_n = \frac{\pi n}{M},$$

results in the following model, usually termed the Dent model or the modified Jakes model:

$$R(t, k) = \sqrt{\frac{2}{M}} \sum_{n=1}^M A_k(n) (\cos \beta_n + j \sin \beta_n) \cos(2\pi f_d t \cos \alpha_n + \theta_n).$$

The weighting functions $A_k(n)$ are the k^{th} Walsh–Hadamard sequence in n . Since these have zero cross-correlation by design, this model results in uncorrelated waveforms. The phases θ_n can be initialised randomly and have no effect on the correlation properties. The [fast Walsh transform](#) can be used to efficiently generate samples using this model.

The Jakes's model also popularised the Doppler spectrum associated with Rayleigh fading, and, as a result, this Doppler spectrum is often termed Jakes's spectrum.

ELC 5396: Digital Communications

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Signaling over Fading Channels

- 1 FIR Modeling of Doubly Spread Channels
- 2 Diversity Techniques
- 3 Space Diversity-on-Receive Systems

FIR Modeling of Doubly Spread Channels

- Impulse response of baseband channel $\tilde{h}(\tau; t)$ and its transfer function $\tilde{H}(f; t)$

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FIR Modeling of Doubly Spread Channels

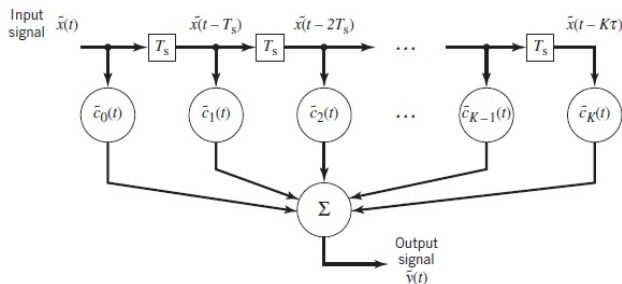
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$$\begin{aligned}y(t) &= \sum_{n=-\infty}^{\infty} x(t - nT_s) \left[\int_{-\infty}^{\infty} h(\tau; t)\text{sinc}(\frac{\tau}{T_s} - n)d\tau \right] \\&= \sum_{n=-\infty}^{\infty} x(t - nT_s)c_n(t)\end{aligned}$$

FIR Modeling of Doubly Spread Channels



- FIR model of a time-varying channel.
- For time-varying Rayleigh fading channels, $c_n(t)$ is zero-mean complex Gaussian process.
- For WSS channels, $E[|c_n(t)|^2] \approx T_s^2 p(nT_s)$, where $p(nT_s)$ is a discrete version of the power-delay profile.

The multipath fading phenomenon as an inherent characteristic of a wireless channel.

Diversity:

If several replicas of the information-bearing signal can be transmitted simultaneously over independently fading channels, then there is a good likelihood that at least one of the received signals will not be severely degraded by channel fading.

Diversity Techniques

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- ③ Space diversity – Several receiving antennas spaced sufficiently far apart (spatial separation should be sufficiently large to reduce correlation between diversity branches, e.g., $> 10\lambda$).

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 - Equalization in a TDM/TDMA system provides similar performance as multipath diversity.

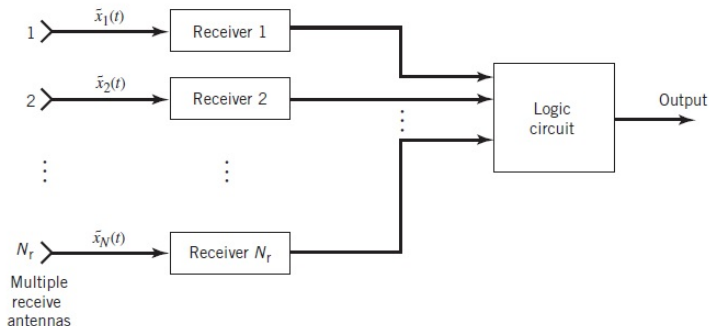
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- 3 Diversity on both transmit and receive – The use of multiple antennas at both the transmitter and receiver.

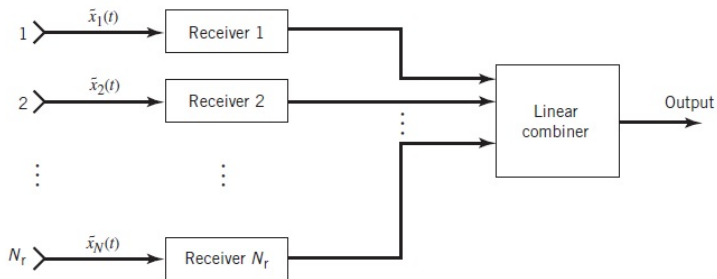
Space Diversity-on-Receive Systems

Selection Combining



Space Diversity-on-Receive Systems

Maximal-Ratio Combining and Equal-Gain Combining



BER vs. SNR in a flat fading channel

In a flat fading channel (or narrowband system), the CIR (channel impulse response) reduces to a single impulse scaled by a time-varying complex coefficient.

The received (equivalent lowpass) signal is of the form

$$r(t) = a(t)e^{j\phi(t)}s(t) + n(t)$$



We assume that the phase changes “slowly” and can be perfectly tracked

=> important for coherent detection

BER vs. SNR (cont.)

We assume:

the time-variant complex channel coefficient changes slowly ($\Rightarrow$ constant during a symbol interval)

the channel coefficient magnitude (= attenuation factor) a is a Rayleigh distributed random variable

coherent detection of a binary PSK signal (assuming ideal phase synchronization)

Let us define instantaneous SNR and average SNR:

$$\gamma = a^2 E_b / N_0 \qquad \gamma_0 = E \{ a^2 \} \cdot E_b / N_0$$

BER vs. SNR (cont.)

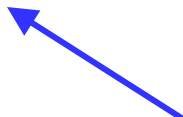
Since

$$p(a) = \frac{2a}{E\{a^2\}} e^{-a^2/E\{a^2\}} \quad a \geq 0,$$

using

$$p(\gamma) = \frac{p(a)}{|d\gamma/da|}$$

Rayleigh distribution



we get

$$p(\gamma) = \frac{1}{\gamma_0} e^{-\gamma/\gamma_0} \quad \gamma \geq 0.$$

Exponential distribution



BER vs. SNR (cont.)

The average bit error probability is

$$P_e = \int_0^{\infty} P_e(\gamma) p(\gamma) d\gamma$$

Important formula
for obtaining
statistical average

where the bit error probability for a certain value of a is

$$P_e(\gamma) = Q\left(\sqrt{2a^2 E_b / N_0}\right) = Q\left(\sqrt{2\gamma}\right).$$

2-PSK

We thus get

$$P_e = \int_0^{\infty} Q\left(\sqrt{2\gamma}\right) \frac{1}{\gamma_0} e^{-\gamma/\gamma_0} d\gamma = \frac{1}{2} \left(1 - \sqrt{\frac{\gamma_0}{1 + \gamma_0}} \right).$$

BER vs. SNR (cont.)

Approximation for large values of average SNR is obtained in the following way. First, we write

$$P_e = \frac{1}{2} \left(1 - \sqrt{\frac{\gamma_0}{1 + \gamma_0}} \right) = \frac{1}{2} \left(1 - \sqrt{1 + \frac{-1}{1 + \gamma_0}} \right)$$

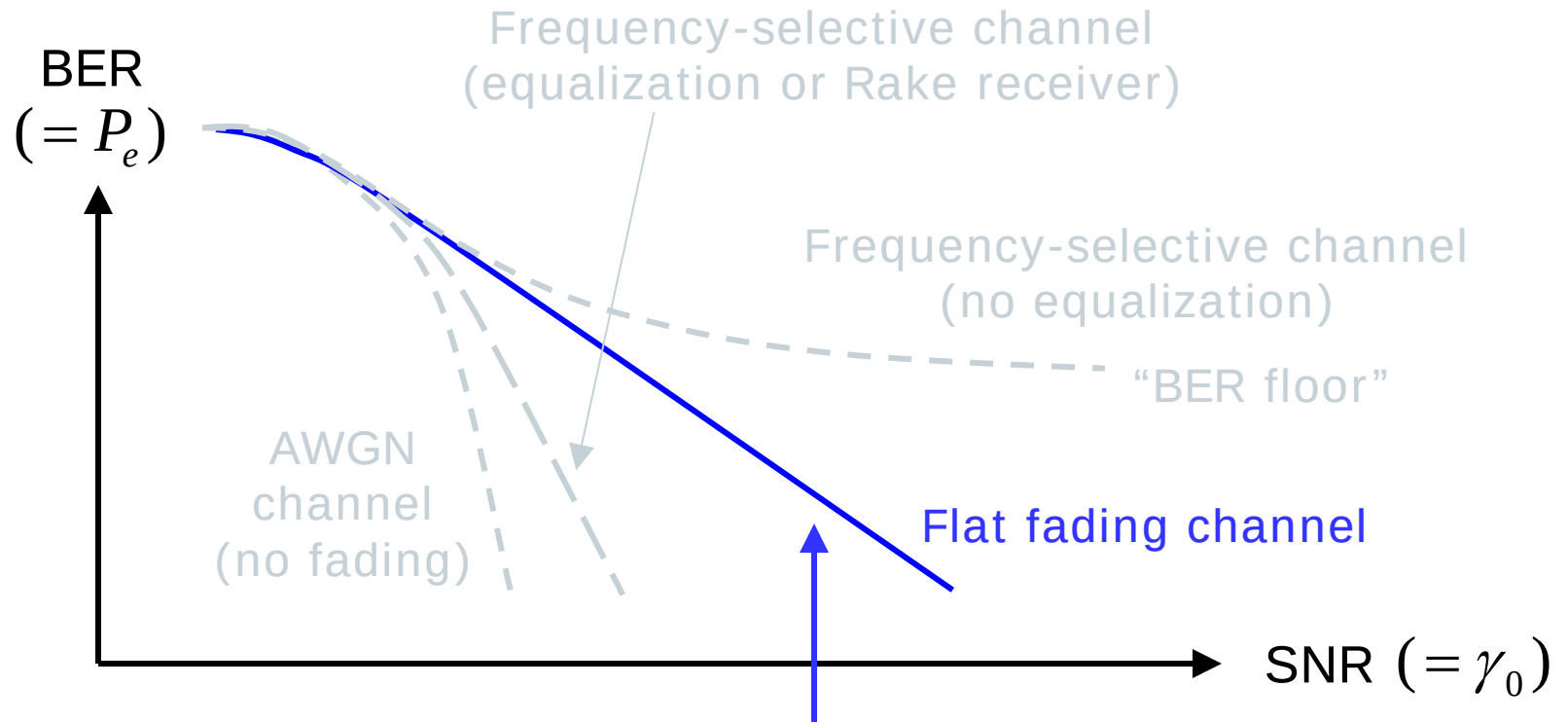
Then, we use

$$\sqrt{1 + x} = 1 + x/2 + \dots$$

which leads to

$$P_e \approx 1/4\gamma_0 \quad \text{for large } \gamma_0 .$$

BER vs. SNR (cont.)



$P_e \approx 1/4\gamma_0$ means a straight line in log/log scale

BER vs. SNR, summary

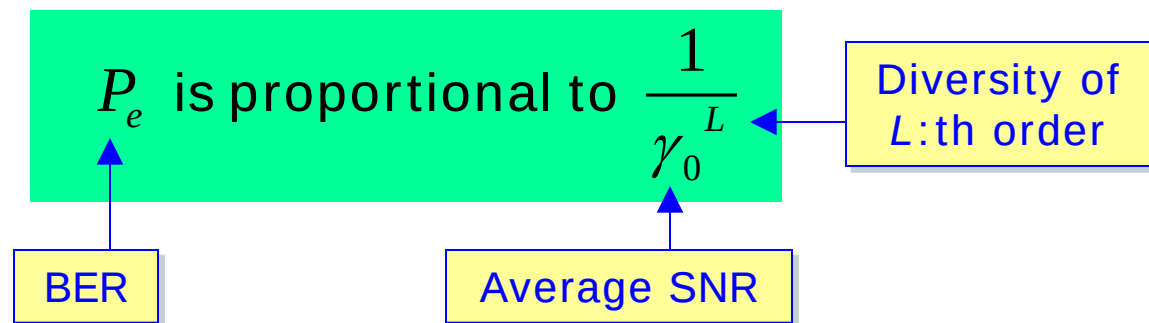
Modulation	$P_e(\gamma)$	P_e	P_e (for large γ_0)
2-PSK	$Q(\sqrt{2\gamma})$	$\frac{1}{2}\left(1 - \sqrt{\frac{\gamma_0}{1 + \gamma_0}}\right)$	$1/4\gamma_0$
DPSK	$e^{-\gamma}/2$	$1/(2\gamma_0 + 2)$	$1/2\gamma_0$
2-FSK (coh.)	$Q(\sqrt{\gamma})$	$\frac{1}{2}\left(1 - \sqrt{\frac{\gamma_0}{2 + \gamma_0}}\right)$	$1/2\gamma_0$
2-FSK (non-c.)	$e^{-\gamma/2}/2$	$1/(\gamma_0 + 2)$	$1/\gamma_0$

Better performance through diversity

Diversity $\Leftrightarrow$ the receiver is provided with multiple copies of the transmitted signal. The multiple signal copies should experience *uncorrelated fading* in the channel.

In this case the probability that *all* signal copies fade simultaneously is reduced dramatically with respect to the probability that a *single* copy experiences a fade.

As a rough rule:



Selection diversity vs. signal combining

Selection diversity: Signal with best quality is selected.

Equal Gain Combining (EGC)

Signal copies are combined coherently:

$$Z_{EGC} = \sum_{i=1}^L a_i e^{j\phi_i} e^{-j\phi_i} = \sum_{i=1}^L a_i$$

Maximum Ratio Combining (MRC, best SNR is achieved)

Signal copies are weighted and combined coherently:

$$Z_{MRC} = \sum_{i=1}^L a_i e^{j\phi_i} a_i e^{-j\phi_i} = \sum_{i=1}^L a_i^2$$

Selection diversity performance

We assume:

- (a) uncorrelated fading in diversity branches
- (b) fading in i :th branch is Rayleigh distributed
- (c) $\Rightarrow$ SNR is exponentially distributed:

$$p(\gamma_i) = \frac{1}{\gamma_0} e^{-\gamma_i/\gamma_0}, \quad \gamma_i \geq 0.$$

PDF

Probability that SNR in branch i is less than threshold y :

$$P(\gamma_i < y) = \int_0^y p(\gamma_i) d\gamma_i = 1 - e^{-y/\gamma_0}.$$

CDF

Selection diversity (cont.)

Probability that SNR in every branch (i.e. all L branches) is less than threshold y :

$$P(\gamma_1, \gamma_2, \dots, \gamma_L < y) = \left[\int_0^y p(\gamma_i) d\gamma_i \right]^L = \left[1 - e^{-y/\gamma_0} \right]^L.$$

Note: this is true only if the fading in different branches is independent (and thus uncorrelated) and we can write

$$p(\gamma_1, \gamma_2, \dots, \gamma_L) = p(\gamma_1) p(\gamma_2) \dots p(\gamma_L).$$

Selection diversity (cont.)

Differentiating the cdf (cumulative distribution function) with respect to y gives the pdf

$$p(y) = L \left[1 - e^{-y/\gamma_0} \right]^{L-1} \cdot \frac{e^{-y/\gamma_0}}{\gamma_0}$$

which can be inserted into the expression for average bit error probability

$$P_e = \int_0^{\infty} P_e(y) p(y) dy.$$

The mathematics is unfortunately quite tedious ...

Selection diversity (cont.)

... but as a general rule, for large γ_0 it can be shown that

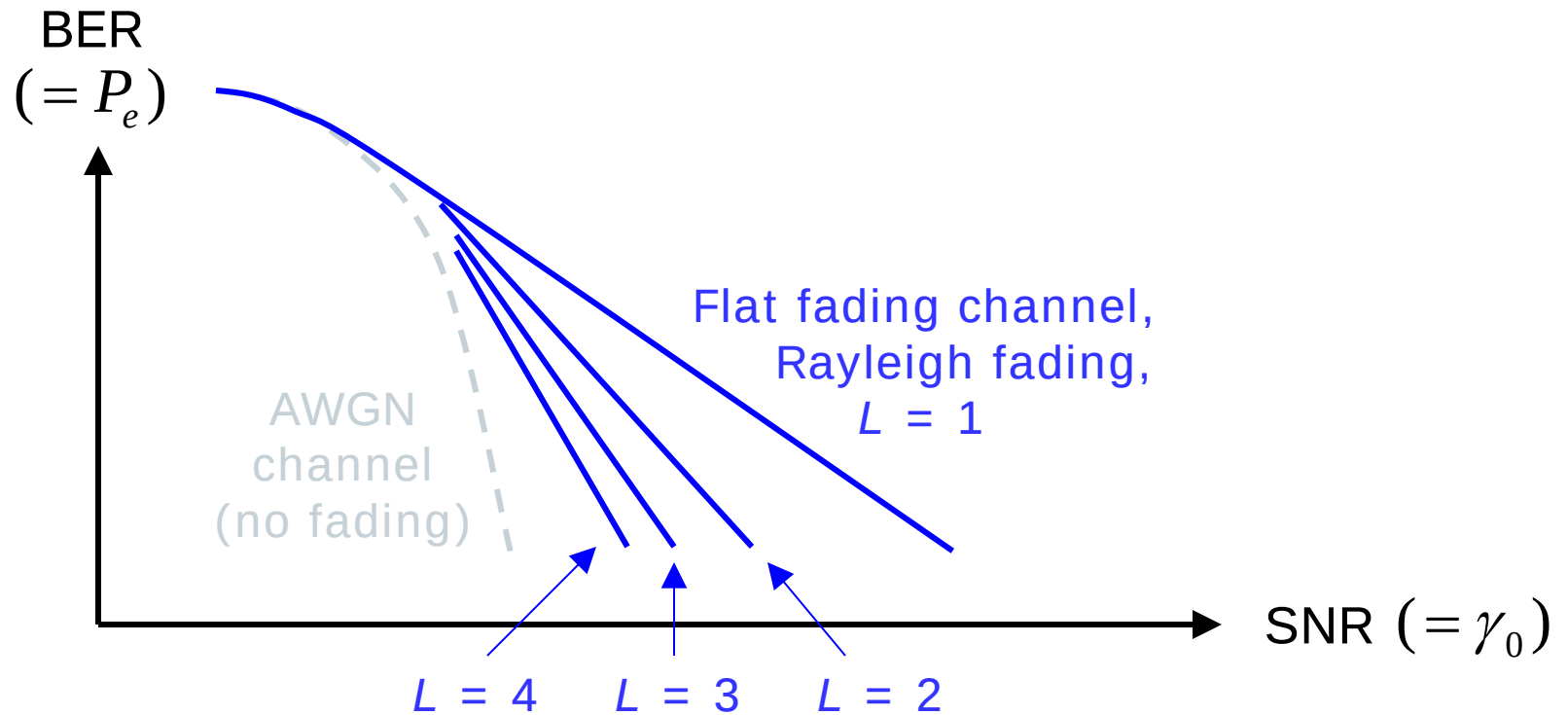
$$P_e \text{ is proportional to } \frac{1}{\gamma_0^L}$$

regardless of modulation scheme (2-PSK, DPSK, 2-FSK).

The largest diversity gain is obtained when moving from $L = 1$ to $L = 2$. The relative increase in diversity gain becomes smaller and smaller when L is further increased.

This behaviour is typical for all diversity techniques.

BER vs. SNR (diversity effect)



MRC performance

Rayleigh fading => SNR in i :th diversity branch is

$$\gamma_i = \frac{E_b}{N_0} a_i^2 = \frac{E_b}{N_0} (x_i^2 + y_i^2)$$

Rayleigh distributed magnitude

Gaussian distributed
quadrature components

In case of L uncorrelated branches with same fading statistics, the MRC output SNR is

$$\gamma = \frac{E_b}{N_0} (a_1^2 + a_2^2 \dots + a_L^2) = \frac{E_b}{N_0} (x_1^2 + y_1^2 \dots + x_L^2 + y_L^2)$$

MRC performance (cont.)

The pdf of γ follows the *chi-square distribution* with $2L$ degrees of freedom

Reduces to exponential pdf when $L = 1$

$$p(\gamma) = \frac{\gamma^{L-1}}{\gamma_0^L \Gamma(L)} e^{-\gamma/\gamma_0} = \frac{\gamma^{L-1}}{\gamma_0^L (L-1)!} e^{-\gamma/\gamma_0}$$

Gamma functionFactorial

For 2-PSK, the average BER is

$$P_e = \left(\frac{1-\mu}{2} \right)^L \sum_{k=0}^{L-1} \binom{L-1+k}{k} \left(\frac{1+\mu}{2} \right)^k$$

$$P_e = \int_0^{\infty} P_e(\gamma) p(\gamma) d\gamma$$

$$P_e(\gamma) = Q(\sqrt{2\gamma})$$

$$\mu = \sqrt{\gamma_0/(1+\gamma_0)}$$

MRC performance (cont.)

For large values of average SNR this expression can be approximated by

$$P_e = \left(\frac{1}{4\gamma_0} \right)^L \binom{2L-1}{L}$$

which again is according to the general rule

$$P_e \text{ is proportional to } \frac{1}{\gamma_0^L}.$$

MRC performance (cont.)

The second term in the BER expression does not increase dramatically with L :

$$\binom{2L-1}{L} = \frac{(2L-1)!}{L! \cdot (L-1)!}$$

$= 1$	$L = 1$
$= 3$	$L = 2$
$= 10$	$L = 3$
$= 35$	$L = 4$

BER vs. SNR for MRC, summary

For large $\gamma_0 \Rightarrow P_e = \left(\frac{1}{k\gamma_0} \right)^L \binom{2L-1}{L}$

Modulation	$P_e(\gamma)$	P_e (for large γ_0)
2-PSK	$Q(\sqrt{2\gamma})$	$k = 4$
DPSK		$k = 2$
2-FSK (coh.)	$Q(\sqrt{\gamma})$	$k = 2$
2-FSK (non-c.)		$k = 1$

Why is MRC optimum performance?

Let us investigate the performance of a signal combining method in general using arbitrary weighting coefficients g_i .

Signal magnitude and noise energy/bit at the output of the combining circuit:

$$Z = \sum_{i=1}^L g_i \cdot a_i \qquad N_t = N_0 \sum_{i=1}^L g_i^2$$

SNR after combining:

$$\gamma = \frac{Z^2 E_b}{N_t} = \frac{E_b \left(\sum g_i a_i \right)^2}{N_0 \sum g_i^2}$$

Why is MRC optimum performance? (cont.)

Applying the Schwarz inequality

$$\left(\sum g_i a_i\right)^2 \leq \sum g_i^2 \sum a_i^2$$

it can be easily shown that in case of equality we must have $g_i = a_i$ which in fact is the definition of MRC.

Thus for MRC the following important rule applies (the rule also applies to SIR = Signal-to-Interference Ratio):

$$\gamma = \sum_{i=1}^L \gamma_i$$

Output SNR or SIR = sum of
branch SNR or SIR values

Matched filter = "full-scale" MRC

Let us consider a single symbol in a narrowband system (without ISI). If the sampled symbol waveform before matched filtering consists of $L+1$ samples

$$r_k, \quad k = 0, 1, 2, \dots, L$$

the impulse response of the matched filter also consists of $L+1$ samples

$$h_k = r_{L-k}^*$$

Definition of matched filter

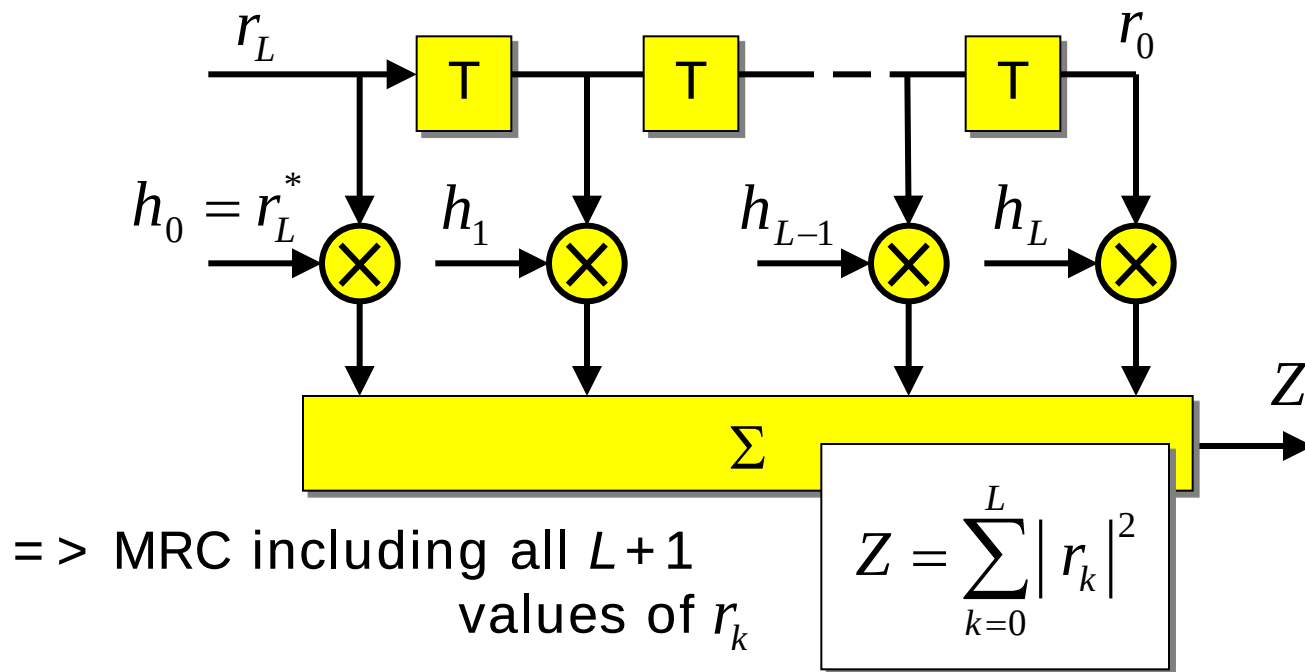
and the output from the matched filter is

MRC !

$$Z = \sum_{k=0}^L h_k r_{L-k} = \sum_{k=0}^L r_{L-k}^* r_{L-k} = \sum_{k=0}^L |r_k|^2$$

Matched filter = MRC (cont.)

The discrete-time (sampled) matched filter can be presented as a transversal FIR filter:



What is MIMO Wireless Technology

MIMO: Multiple Input Multiple Output technology uses multiple antennas to make use of reflected signals to provide gains in channel robustness and throughput.

Multiple-input multiple-output, or MIMO, is a radio communications technology or RF technology that is being mentioned and used in many new technologies these days.

Wi-Fi, LTE; Long Term Evolution, and many other radio, wireless and RF technologies are using the new MIMO wireless technology to provide increased link capacity and spectral efficiency combined with improved link reliability using what were previously seen as interference paths.

Even now many there are many MIMO wireless routers on the market, and as this RF technology is becoming more widespread, more MIMO routers and other items of wireless MIMO equipment will be seen.



Typical modern WiFi router using MIMO technology with multiple antennas

As the technology is complex many engineers are asking what is MIMO and how does it work.

MIMO development and history

MIMO technology has been developed over many years. Not only did the basic MIMO concepts need to be formulated, but in addition to this, new technologies needed to be developed to enable MIMO to be fully implemented. New levels of processing were needed to allow some of the features of spatial multiplexing as well as to utilise some of the gains of spatial diversity.

Up until the 1990s, spatial diversity was often limited to systems that switched between two antennas or combined the signals to provide the best signal. Also various forms of beam switching were implemented, but in view of the levels of processing involved and the degrees of processing available, the systems were generally relatively limited.

However with the additional levels of processing power that started to become available, it was possible to utilise both spatial diversity and full spatial multiplexing.

The initial work on MIMO systems focussed on basic spatial diversity - here the MIMO system was used to limit the degradation caused by multipath propagation. However this was only the first step as system then started to utilise the multipath propagation to advantage, turning

the additional signal paths into what might effectively be considered as additional channels to carry additional data.

Two researchers: Arogyaswami Paulraj and Thomas Kailath were first to propose the use of spatial multiplexing using MIMO in 1993 and in the following year their US patent was granted.

However it fell to Bell Labs to be the first to demonstrate a laboratory prototype of spatial multiplexing in 1998.

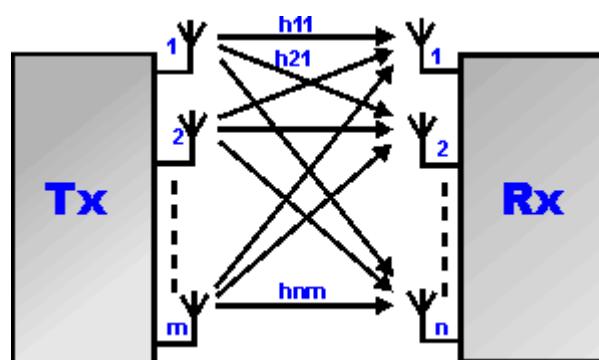
MIMO -Multiple Input Multiple Output basics

A channel may be affected by fading and this will impact the signal to noise ratio. In turn this will impact the error rate, assuming digital data is being transmitted. The principle of diversity is to provide the receiver with multiple versions of the same signal. If these can be made to be affected in different ways by the signal path, the probability that they will all be affected at the same time is considerably reduced. Accordingly, diversity helps to stabilise a link and improves performance, reducing error rate.

Several different diversity modes are available and provide a number of advantages:

- **Time diversity:** Using time diversity, a message may be transmitted at different times, e.g. using different timeslots and channel coding.
- **Frequency diversity:** This form of diversity uses different frequencies. It may be in the form of using different channels, or technologies such as spread spectrum / OFDM.
- **Space diversity :** Space diversity used in the broadest sense of the definition is used as the basis for MIMO. It uses antennas located in different positions to take advantage of the different radio paths that exist in a typical terrestrial environment.

MIMO is effectively a radio antenna technology as it uses multiple antennas at the transmitter and receiver to enable a variety of signal paths to carry the data, choosing separate paths for each antenna to enable multiple signal paths to be used.



General Outline of MIMO system

One of the core ideas behind MIMO wireless systems space-time signal processing in which time (the natural dimension of digital communication data) is complemented with the spatial dimension inherent in the use of multiple spatially distributed antennas, i.e. the use of multiple antennas located at different points. Accordingly MIMO wireless systems can be viewed as a logical extension to the smart antennas that have been used for many years to improve wireless.

It is found between a transmitter and a receiver, the signal can take many paths. Additionally by moving the antennas even a small distance the paths used will change. The variety of paths available occurs as a result of the number of objects that appear to the side or even in the direct path between the transmitter and receiver. Previously these multiple paths only served to introduce interference. By using MIMO, these additional paths can be used to advantage. They can be used to provide additional robustness to the radio link by improving the signal to noise ratio, or by increasing the link data capacity.

The two main formats for MIMO are given below:

- ***Spatial diversity:*** Spatial diversity used in this narrower sense often refers to transmit and receive diversity. These two methodologies are used to provide improvements in the signal to noise ratio and they are characterised by improving the reliability of the system with respect to the various forms of fading.
- ***Spatial multiplexing :*** This form of MIMO is used to provide additional data capacity by utilising the different paths to carry additional traffic, i.e. increasing the data throughput capability.

As a result of the use multiple antennas, MIMO wireless technology is able to considerably increase the capacity of a given channel while still obeying Shannon's law. By increasing the number of receive and transmit antennas it is possible to linearly increase the throughput of the channel with every pair of antennas added to the system. This makes MIMO wireless technology one of the most important wireless techniques to be employed in recent years. As spectral bandwidth is becoming an ever more valuable commodity for radio communications systems, techniques are needed to use the available bandwidth more effectively. MIMO wireless technology is one of these techniques.

OFDM

[Orthogonal frequency-division multiplexing - Wikipedia](#)

Prepare short notes from the Wikipedia page.

5.10 Spread Spectrum Modulation Techniques

All of the modulation and demodulation techniques described so far strive to achieve greater power and/or bandwidth efficiency in a stationary additive white Gaussian noise channel. Since bandwidth is a limited resource, one of the primary design objectives of all the modulation schemes detailed thus far is to minimize the required transmission bandwidth. Spread spectrum techniques, on the other hand, employ a transmission bandwidth that is several orders of magnitude *greater* than the minimum required signal bandwidth. While this system is very bandwidth inefficient for a single user, the advantage of spread spectrum is that many users can simultaneously use the same bandwidth without significantly interfering with one another. In a multiple-user, *multiple access interference* (MAI) environment, spread spectrum systems become very bandwidth efficient.

Apart from occupying a very large bandwidth, spread spectrum signals are *pseudorandom* and have noise-like properties when compared with the digital information data. The spreading waveform is controlled by a *pseudo-noise (PN) sequence* or *pseudo-noise code*, which is a binary sequence that appears random but can be reproduced in a deterministic manner by intended receivers. Spread spectrum signals are demodulated at the receiver through crosscorrelation with a locally-generated version of the pseudorandom carrier. Crosscorrelation with the correct PN sequence *despreads* the spread spectrum signal and restores the modulated message in the same narrow band as the original data, whereas crosscorrelating the signal from an undesired user results in a very small amount of wideband noise at the receiver output.

Spread spectrum modulation has many properties that make it particularly well-suited for use in the mobile radio environment. The most important advantage is its inherent interference rejection capability. Since each user is assigned a unique PN code which is approximately orthogonal to the codes of other users, the receiver can separate each user based on their codes, even though they occupy the same spectrum at all times. This implies that, up to a certain number of users, interference between spread spectrum signals using the same frequency is negligible. Not only can a particular spread spectrum signal be recovered from a number of other spread spectrum signals, it is also possible to completely recover a spread spectrum signal even when it is jammed by a narrowband interferer. Since narrowband interference affects only a small portion of the spread spectrum signal, it can easily be removed through notch filtering without much loss of information. Since all users are able to share the same spectrum, spread spectrum may eliminate frequency planning, since all cells can use the same channels.

Resistance to multipath fading is another fundamental reason for considering spread spectrum systems for wireless communications. Chapter 4 showed that wideband signals are frequency selective. Since spread spectrum signals

have uniform energy over a very large bandwidth, at any given time only a small portion of the spectrum will undergo fading (recall the comparison between wideband and narrowband signal responses in multipath channels in Chapter 4). Viewed in the time domain, the multipath resistance properties are due to the fact that the delayed versions of the transmitted PN signal will have poor correlation with the original PN sequence, and will thus appear as another uncorrelated user which is ignored by the receiver. Spread spectrum systems are not only resistant to multipath fading, but they can also exploit the multipath components to improve the performance of the system. This can be done using a RAKE receiver which combines the information obtained from several resolvable multipath components. A RAKE receiver consists of a bank of correlators, each of which correlate to a particular multipath component of the desired signal. The correlator outputs may be weighted according to their relative strengths and summed to obtain the final signal estimate [Pri58]. RAKE receivers are described in Chapter 6.